

# An Atmospheric Energy Balance Model for coupling to MOM

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## Abstract

We describe herein the equations and numerical methods used for an atmospheric Energy Balance Model that could be coupled to MOM, and sea-ice and land-surface models, in order to provide a simple yet relevant boundary condition for climate experiments. Using the EBM could be an alternative both to restoring boundary conditions (frequently used in ‘ocean-only’ simulations) and to a full atmospheric GCM.

## 1 Introduction

An Energy Balance Model (EBM hereafter) is a simple approximation of the atmosphere considered as a one-layer where transport of heat is usually parameterized as an eddy diffusion law. The atmospheric temperature then changes according to the balance between radiative fluxes, sensible and latent heat exchange at the surface and horizontal diffusion. Our primary goal is to provide a boundary condition for an ocean model such as MOM, that might be used under a variety of circumstances. We want to keep the parameterizations as simple as possible and physically based, not necessarily exquisitely tuned to present day climate since they should be able to represent climates of other times.

The equation of motion is:

$$C_A \frac{dT_A}{dt} = (S + IR) + \text{Lateral Transport} + \text{Exchange with surface} \quad (1)$$

or, term-by-term,

$$C_A \frac{dT_A}{dt} = Q_{SSWA} - Q_{LWAup} - Q_{LWAdown} + Q_{LWOut} + Q_{SH} + Q_{LHA} + C_A \nabla \cdot (K_A \nabla T_A) \quad (2)$$

where the various terms are as follows:

$C_A = \rho_A C_p H_A$  is the heat capacity of the atmospheric column ( $10^7 \text{ J m}^{-2} \text{ K}^{-1}$ ),  $\rho_A$  the air density ( $1.225 \text{ kg m}^{-3}$ ),  $H_A$  the atmosphere height (8320 m), and  $C_p$  the atmospheric specific heat capacity

( $1004 \text{ J kg}^{-1} \text{ K}^{-1}$ );

$T_A$  is the atmosphere temperature [ $^{\circ}\text{C}$ ];

$Q_{SSWA}$  is the short wave absorption in the atmosphere,

$Q_{LWAup}$  the long wave emission to space;

$Q_{LWAdown}$  the long wave emission downward (to the ground or the ocean);

$Q_{LWOup}$  the long wave emission by the ocean or the ground upward;

$Q_{SH}$  the upward sensible heat flux from ocean to atmosphere;

$Q_{LHA}$  the latent heat released by precipitation.

The total heat flux entering the ocean is given by

$$Q_{HO} = Q_{SSWO} + Q_{LWAdown} - Q_{LWOup} - Q_{SH} - Q_{LHO} \quad (3)$$

where  $Q_{SSWO}$  is the downwards shortwave radiation at the surface, and  $Q_{LHO}$  is the latent heat absorbed by evaporation. In all the expressions above, terms involving latent heat or moisture flux are currently set to zero.

Typically, an energy balance in the atmosphere occurs on time scales of the order of a week. This is faster than most ocean-dynamics timescales, and so for ocean modeling purposes one might reasonably suppose that the atmosphere is in equilibrium and neglect the time-derivative on the left-hand-side. However, we keep it for generality. Furthermore, it is numerically easier to time-step the atmosphere, rather than solve an elliptic equation for the temperature, although for explicit methods the atmospheric timestep must then typically be shorter than that of non-eddy-resolving ocean models.

## 2 Radiation

### 2.1 Short waves

#### 2.1.1 Incoming short waves

The incoming short wave at the top of the atmosphere depends on latitude, not only because of the projection of the disc of incoming solar beams on the sphere but also because of the tilt of the rotation axis of the Earth and the ellipticity of its trajectory around the sun. On annual mean, simple formulations have been used, like  $Q_{SSW} = S_0/4 \times (1 + 0.7(\cos^2(\vartheta) - 2/3))$ , where  $S_0$  is the solar constant ( $1360 \text{ W m}^{-2}$ ) and  $\vartheta$  is latitude. Sellers (1969) tabulated this latitudinal distribution, and more recently Chylek and Coakley (1975). This is what we use.

#### 2.1.2 Albedo

The albedo at the top of the atmosphere ( $\alpha$ ) is shown to vary widely from equator to pole (0.2 to 0.6): that might be the influence of the snow/ice covers at high latitude, but it is mainly a

consequence of the increased reflection of solar beams entering the atmosphere with decreasing incidence (Ref?). Graves *et al.* (1993) provide a relationship for the annual mean background albedo for average sky as a function of latitude (excluding the effect of increased albedo on snow) that differs from previously used parameterizations ( $0.25 + 0.18x^2$ , Hyde *et al.* 1990) mainly by an increased latitudinal variation with latitude and a slight asymmetry between the hemispheres:  $\alpha = 0.679 - 0.012P_1(x) - 0.241P_2(x)$ , where  $x = \sin(\vartheta)$  and  $P_n$  is the  $n$ th order Legendre polynomial:  $P_1(x) = x$  and  $P_2(x) = \frac{1}{2}(3x^2 - 1)$ . Graves *et al.* suggest  $\alpha = 0.2005 + 0.012x + 0.3615x^2$ . We use the simpler symmetric formula:

$$\alpha = 0.20 + 0.36 \sin^2(\vartheta). \quad (4)$$

Note that this is the atmospheric albedo only, and does not include any surface effects.

### 2.1.3 Surface albedo

The surface albedo  $\alpha_s$  depends on the type of the surface, whether it is land or ocean. In the simplest version of the model, the surface albedos of the land and ocean are set in namelists and are constant, and ice is not present. This will change when sea-ice and land-surface models are used.

### 2.1.4 Absorption in the atmosphere

Absorption of solar radiation in the atmosphere is calculated using a simple adding method (e.g. Lacis and Hansen) that takes into account multiple reflections between land and atmosphere. This is done algebraically, so there is little computational cost. We suppose the absorption in the atmosphere,  $C_0$ , is constant, (e.g.,  $C_0 = 0.3$ .) Then, if there is a unit flux of solar radiation is incident at the top of the atmosphere, a fraction  $\alpha$  is reflected and  $(1 - \alpha)$  is the net flux entering the TOA. An amount  $[(1 - \alpha)C_0]$  is absorbed in the atmosphere,  $[(1 - \alpha)(1 - C_0)]$  hits the ground,  $[(1 - \alpha)(1 - C_0)\alpha_s]$  is reflected to the atmosphere, and so on.

Summing the multiple reflections and absorptions, the total short wave absorption in the atmosphere is:

$$Q_{SSWA} = Q_{SSW} \frac{(1 - \alpha)C_0(1 + (1 - C_0)\alpha_s)}{1 - \alpha(1 - C_0)^2\alpha_s}$$

where  $\alpha_s$  is the surface albedo and  $Q_{SSW}$  is the incoming solar radiation at the top of the atmosphere.

The short wave absorption in the ground or the ocean after multiple reflection is:

$$Q_{SSWO} = Q_{SSW} \frac{(1 - \alpha)(1 - C_0)(1 - \alpha_s)}{1 - \alpha(1 - C_0)^2\alpha_s}$$

The flux reflected back to space is

$$Q_{SSWR} = Q_{SSW} \left( \alpha + \frac{(1 - \alpha)^2(1 - C_0)^2\alpha_s}{1 - \alpha(1 - C_0)^2\alpha_s} \right)$$

It may be verified that  $Q_{SSWA} + Q_{SSWO} + Q_{SSWR} = Q_{SSW}$ .

It is a simple matter for the sea-ice model to provide albedos for ice and ocean, and the land-surface model to provide land albedo.

## 2.2 Long waves

### 2.2.1 Top of the atmosphere

Various parametrizations are used in the literature, basically of 2 types: one based on a radiation law with specific atmospheric emissivity (and so proportional to  $\sigma T^4$ , and the other based on a linearization, as in by Budyko (1969). Our code allows for both. Thus:

$$Q_{LW\text{Aup}} = A_{\text{Aup}} + B_{\text{Aup}}T_a + \epsilon_{\text{Aup}}\sigma(273.15 + T_a)^4 + (\Gamma - 1)Q_{LW\text{Oup}}$$

where  $\Gamma$  is the absorption of long wave emitted by the ground/ocean in the atmosphere and  $Q_{LW\text{Oup}}$  is the radiation emitted at the surface. Temperature is in Celsius. Thus far we have set  $\Gamma = 1$ . In the code one would normally choose *either* the linear form  $A_{\text{Aup}} + B_{\text{Aup}}T_a$  *or* the emissivity form  $\epsilon_{\text{Aup}}\sigma(273.15 + T_a)^4$ . Default values are  $A_{\text{Aup}} = 210 \text{ W m}^{-2}$ ,  $B_{\text{Aup}} = 2 \text{ W m}^{-2} \text{ K}^{-1}$  and  $\epsilon_{\text{Aup}} = 0$ .

### 2.2.2 Surface

At the surface, we have both the atmospheric emission downward and the ground/ocean emission upward. The atmospheric emission may be linearized or follow a back body emission law. The radiation emitted by the surface follows a black-body law, with a specific emissivity of 0.86 for the ocean, but this might change on availability of linear coefficients. Thus

$$Q_{LW\text{Adown}} = A_{\text{Adown}} + B_{\text{Adown}}T_a + T_a\epsilon_{\text{Ad}}\sigma(273.15 + T_a)^4, \quad \epsilon_{\text{Ad}} = 0.85$$

$$Q_{LW\text{Oup}} = \epsilon_O\sigma(273.15 + T_O)^4, \quad \epsilon_O = 0.96$$

$$Q_{LW\text{Lup}} = \epsilon_L\sigma(273.15 + T_G)^4, \quad \epsilon_L = 0.90$$

where  $\sigma$  is the Stefan-Boltzman constant ( $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-1}$ ). Currently, the default values of  $A_{\text{Adown}}$  and  $B_{\text{Adown}}$  are zero.

## 3 Surface Temperature and Land

The surface temperature of the ocean is assumed to be provided by the ocean model. The EBM then calculates the appropriate flux terms and uses these to step forward the EBM by one timestep. It also provides the fluxes for the ocean model. If ice is present it may be treated in exactly the same way; the ice model provides a surface temperature and the EBM calculates the fluxes.

In the absence of a land model that provides land temperatures and albedos in the same way, the EBM itself be used to provide a simple land model. The assumption is made that the heat capacity

of the land is zero. The land surface then is in equilibrium with the radiative and sensible heat fluxes at the surface. The EBM then proceeds simply by setting to zero all the fluxes from the land to the atmosphere. The EBM need not calculate the land temperature, although it could be diagnosed if needed (say to calculate a surface albedo).

## 4 Atmospheric heat transport

The atmospheric heat transport is assumed to follow a two-dimensional eddy diffusion law. The tuning of this eddy diffusivity with latitude to obtain a reasonable Earth climatology was performed extensively in the 1970's and implied large variations between the tropics and the poles, by a factor of 2 to 3. Our eddy diffusion may have a specified latitudinal variation, although often will simply be a constant. A typical value for the diffusion coefficient is  $K_A = 10^6 \text{ m}^2 \text{ s}^{-1}$ , such that  $D = K_A C_A / a^2 = 0.252 \text{ W m}^{-2} \text{ K}^{-1}$ , where  $a$  is the Earth radius (6371 km).

The zonal boundaries (at the Northern and Southern ends of the domain) have a no-flux condition. The meridional boundaries may be either no-flux or periodic. The former is appropriate if the integration is over a limited domain, the latter if the domain goes completely around a latitude circle. The grid is latitude-longitude. It need not be uniform, but thus far we have only integrated it using a uniform grid (typically  $\Delta\vartheta = \Delta\lambda = 2^\circ$ ).

### 4.1 Advection?

The advection of  $T_A$  by the wind field, imposed or calculated, might be considered. A simple scaling analysis shows it plays a significant role in the balance, as one could expect from wind blowing from land to the ocean in the winter hemisphere. Uncertainties result from the unknown profile of temperature and wind with height. However it might have an important contribution to removing the isotropy of the diffusion operator.

## 5 Sensible heat

The sensible heat flux is simply related to the difference of atmospheric and oceanic temperature. In bulk formulas, the wind speed is taken into account in the exchange coefficient. In the absence of wind field, this coefficient is taken as a constant, whose influence on the fluxes is large. For a typical wind speed of 10 m/s, the bulk formula used by Fanning and Weaver (1996) gives a value of  $15 \text{ W m}^{-2} \text{ K}^{-1}$ . That is our default value. If  $T_{AS}$  is the atmospheric temperature at 10 m then the exchange of sensible heat is parameterized as

$$Q_{SH} = \gamma(T_O - T_{AS}) \quad (5)$$

In general, the model predicted temperature,  $T_A$ , is not the temperature at 10 m above the surface.

It may be the temperature at 500 mb, for example. In general therefore, (5) becomes:

$$Q_{SH} = \gamma(T_O - (T_A + \Delta T)) \quad (6)$$

where, if  $T_A$  is the 500mb temperature,  $\Delta T$  might typically be about 20 K. It is set in a namelist.

## 6 Moisture

Not considered yet, although the results of the ‘Energy Moisture Balance Model’ of Fanning and Weaver (1996) are apparently encouraging. Thus far, the energy balance model has been run assuming all the heat transport is by sensible heat or radiation. When coupled to an ocean model that needs an E–P field, this must be specified. In the anomaly model a diffusive scheme for moisture would seem sensible.

Similarly, latent heat release could be implemented if the evaporation (E) and/or precipitation (P) fields are provided or calculated with a moisture model. Then

$$Q_{LHA} = \rho_0 L_V P \quad \text{and} \quad Q_{LHo} = \rho_0 L_V E$$

where P and E are in m/s,  $\rho_0$  the density of water ( $1000 \text{ kg m}^{-3}$ ) and  $L_V$  is the latent heat of evaporation ( $2.5 \times 10^6 \text{ J kg}^{-1}$ ).

## 7 Numerics

Two different time stepping scheme have been implemented so far.

- (i) A simple Forward-in-time, centered-in-space scheme which is limited in time step by the diffusion operator:  $dt < (dx^2)/(2K_A)$ .
- (ii) A DuFort-Frankel scheme which is unconditionally stable, yet it may be cast in a purely explicit form without the need for an elliptic solve. Although the cost per timestep of the EBM is negligible at low resolution, the limitation of the time step by diffusion (as compared to advection in the ocean model) makes it much more expensive at high resolution. A Dufort-Frankel scheme treats the center gridpoint in the Laplacian operator implicitly, but the surrounding terms explicitly. The differencing scheme can be re-arranged to be purely explicit form. We have generally found the Dufort-Frankel method to be preferable, since the forward scheme seems to demand very small timesteps.

Spatial differencing uses a standard five-point stencil for the diffusive term. Using a Dufort-Frankel scheme complicates the coding of this, but does not make the stencil any larger. The differencing is done on a latitude-longitude grid.

Even with Dufort-Frankel, the timesteps of the atmospheric model will frequently be shorter than those of the ocean model. In its current version (even without the flux coupler) the EBM does allow for different timesteps.

## 8 An Atmospheric Temperature Anomaly Model

Recall that the equations of motion are:

$$C_a \frac{\partial T_a}{\partial t} = (S + \text{IR}) + C_a \kappa \nabla^2 T_a + \gamma_T (T_o - (T_a + \Delta T)) \quad (7)$$

where  $T_a$  is the temperature of the atmosphere, a function of both  $x$  and  $y$ ,  $C_a$  is a heat capacity and  $\kappa$  a diffusion coefficient.  $S$  and  $\text{IR}$  are the solar and infra-red fluxes, and the last term represents the exchange of heat with the surface (ocean or land). The parameter  $\gamma_T$  may be chosen to be consistent with  $\lambda_T$ .  $\text{IR}$  depends on  $T_a$  in a more-or-less simple way. One could directly couple such a model to an ocean model to provide surface fluxes. However, given the uncertainty in the form and the details of this model, it is likely that the ensuing fluxes would differ greatly from those observed, leading to an unrealistic ocean circulation. So let us construct a model for atmospheric *anomalies*.

Write  $T_a = \hat{T}_a + T'_a$  and  $T_o = \hat{T}_o + T'_o$ . Then (7) becomes

$$\begin{aligned} C_a \frac{\partial T'_a}{\partial t} &= S - \hat{\text{IR}} + \gamma_T (\hat{T}_o - \hat{T}_a) + C_a \kappa \nabla^2 \hat{T}_a \\ &+ \text{IR}' + \gamma_T (T'_o - T'_a) + C_a \kappa \nabla^2 T'_a \end{aligned} \quad (8)$$

Now assume that the top line on the right-hand-side is zero. Further, note the outgoing atmospheric infra-red radiation is approximated by  $A + BT$  where  $B \sim 2\text{W m}^{-2}\text{C}^{-1}$ . This is much smaller than the surface sensible heat exchange coefficients ( $\gamma_T$ ) which are usually equivalent to values  $> 10\text{W m}^{-2}\text{C}^{-1}$  (cf. Haney's (1971) canonical value of  $35\text{W m}^{-2}\text{C}^{-1}$ ). Thus, ignoring the IR anomaly, (8) reduces to a model for the atmospheric temperature anomaly:

$$\boxed{C_a \frac{\partial T'_a}{\partial t} = C_a \kappa \nabla^2 T'_a + \gamma_T (T'_o - T'_a)} \quad (9)$$

If the response of the atmosphere is sufficiently fast that it is in a local thermodynamic equilibrium with the ocean, this becomes:

$$\boxed{\kappa \nabla^2 T'_a - \frac{\gamma_T}{C_a} T'_a = -\frac{\gamma_T}{C_a} T'_o} \quad (10)$$

Thus, there is no atmospheric temperature anomaly in the absence of an SST anomaly. Also, in the absence of horizontal diffusion (or when integrated over the domain so the diffusion term vanishes) the atmospheric temperature anomaly equals the oceanic temperature anomaly. This would not hold if IR anomalies are included.

Similar equations can be posited for atmospheric moisture anomaly, to wit:

$$\boxed{\frac{\partial q'_a}{\partial t} = \kappa \nabla^2 q'_a + \gamma_q (q'_o - q'_a)} \quad (11)$$

or

$$\kappa \nabla^2 q'_a - \gamma_q q'_a = -\gamma_q q'_o \quad (12)$$

where  $q_a$  is the atmospheric water vapour mixing ratio and  $q_o$  is the saturation mixing ratio at the temperature of the ocean surface. Thus,  $q'_o$  is an anomaly due to a temperature anomaly only.

### 8.1 Use as a boundary condition for ocean model

The above atmospheric temperature anomaly model can be used in conjunction with observed fluxes. The sensible heat flux between atmosphere and ocean is given by

$$F_Q = (\text{Observed Surface Heat Flux}) - \lambda_T(T'_o - T'_a) \quad (13)$$

where  $T'_a$  is to be calculated using (9) or (10). The effects of the anomalous flux of latent heat due to an oceanic temperature anomaly could be explicitly included in (13), but it may be unnecessary (?) since for small anomalies this effectively amounts to a small temperature dependent change in  $\lambda_T$ .

The anomalous evaporative flux should be calculated, though, as it does affect the salinity. Thus we may write

$$F_M = (\text{Observed Precip and Evaporative Flux}) - \lambda_q(q'_o - q'_a) \quad (14)$$

where  $q'_a$  is calculated using (11) or (12). However, this does not allow any direct correction of salinity. Such a correction may be necessary because the observed precipitation fluxes are poorly known, and because of model deficiencies. If so, one may include an additional ‘salinity relaxation’ term in the upper layer model equation. The effective moisture flux becomes:

$$F_M = (\text{Observed Precip and Evaporative Flux}) - \lambda_q(q'_o - q'_a) - \lambda_S(S_o - \hat{S}_o). \quad (15)$$

(Note that  $\lambda_q$  and  $\lambda_S$  have different units, and that  $\lambda_S$  can be set to zero if needed.)

### 8.2 Qualitative properties of the scheme

Suppose the atmosphere responds quickly to the ocean, and consider an oceanic temperature anomaly. This produces an atmospheric anomaly by (10). If the anomaly is on a sufficiently large-scale, then  $T'_a \approx T'_o$ . There is only a small correction term in (13) and the anomaly takes a long time to dissipate. If the whole ocean is anomalously warm, the flux correction will have no effect at all.

If the anomaly is on a small scale, then from (10)  $T'_a \ll T'_o$  and the anomaly will dissipate quickly.

If the ocean is anomalously warm, then using (14) or (15) there will be enhanced evaporation, consistent with the enhanced heat flux, and the ocean will become locally more salty.

If the ocean is anomalously salty (but at the observed temperature), then if (14) is used as the surface boundary condition the moisture flux will be unaffected, as is physically reasonable. On the



other hand if (15) is used then the salt anomaly will relax back to the observed value. Thus, whether salt relaxation is invoked should be a conscious choice on the part of the modeller.

The scheme is similar, but not identical, to one proposed by Willebrand and Rahmstorf. However, the scheme described above seems to have sensible behavior at both large and small scales.

### 8.3 Configuring the code

Two steps must be taken to use the EBM in anomaly mode:

1. Set to zero the solar flux and the mean IR flux (the  $A_{up}$  and similar terms).
2. Include the pre-specified (e.g., climatological) fluxes between surface and atmosphere in the forcing of the surface.

The temperature predicted by the EBM is then the atmospheric anomaly, which is forced only by the oceanic temperature anomalies. A slightly more complicated procedure might be needed if one wished to allow for the effects of temperature anomalies on the solar radiation scheme. One could, with more reason, presumably also include the effects of temperature anomalies in the IR scheme. Note that then the spatial-average atmospheric anomaly would not equal the spatial-average oceanic anomaly, even in steady-state.

## 9 Final Remarks

There should be no difficulty, in principle, with using the EBM in conjunction with a flux-coupler under FMS. The land-surface and sea-ice models provide all the surface properties (temperature, albedo etc) to the EBM. The fluxes are calculated in the same way as before. See the document 'Coupling and EBM to MOM and SIS' by H. Gildor.

Finally, note that the interpretation of the temperature within the EBM is up to the user, and determined by the parameters chosen to exchange heat and radiation between ocean and atmosphere.

## References

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| Parameter       | Description                      | Value                                                     | Source                          |
|-----------------|----------------------------------|-----------------------------------------------------------|---------------------------------|
| $a$             | radius of the Earth              | 6371 km                                                   | NA                              |
| $A$             | linearized long wave coefficient | 210 W m <sup>-2</sup>                                     | Toggweiler <i>et al.</i> (1998) |
| $B$             | linearized long wave coefficient | 2 W m <sup>-2</sup> °C <sup>-1</sup>                      | Toggweiler <i>et al.</i> (1998) |
| $A_G$           |                                  |                                                           |                                 |
| $B_G$           |                                  |                                                           |                                 |
| $C_A$           | atmospheric column heat capacity | 10 <sup>7</sup> J m <sup>-2</sup> K <sup>-1</sup>         |                                 |
| $C_{p_A}$       | specific heat capacity of air    | 1004 J kg <sup>-1</sup> K <sup>-1</sup>                   |                                 |
| $H_A$           | atmosphere height 8320 m         |                                                           |                                 |
| $L_V$           | latent heat of evaporation       | 2.5 × 10 <sup>6</sup> J kg <sup>-1</sup>                  |                                 |
| $S_0$           | solar constant                   | 1360 W m <sup>-2</sup>                                    |                                 |
| $\alpha$        | albedo                           | 0.2 + 0.36 sin <sup>2</sup> ( $\vartheta$ )               | Graves <i>et al.</i> (1993)     |
| $\alpha_S$      | surface albedo                   | 0.06 ocean, 0.2 land                                      |                                 |
| $\epsilon_{Ad}$ | atmospheric emissivity downward  | 0.85                                                      |                                 |
| $\epsilon_L$    | land emissivity                  | 0.90                                                      |                                 |
| $\epsilon_O$    | oceanic emissivity               | 0.96                                                      |                                 |
| $\Gamma$        | atmos. infrared absorption coef. | 1                                                         |                                 |
| $\rho_0$        | water density                    | 1000 kg m <sup>-3</sup>                                   |                                 |
| $\rho_A$        | air density                      | 1.225 kg m <sup>-3</sup>                                  |                                 |
| $\sigma$        | Stefan-Boltzman constant         | 5.67 × 10 <sup>-8</sup> W m <sup>-2</sup> K <sup>-4</sup> |                                 |

Table 1: Summary of model parameters

| Short         | Description                                                 | Parameterization                                                                                  |
|---------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| $Q_{HO}$      | Total heat flux into the ocean                              | $Q_{SSWO} + Q_{LWAdown} - Q_{LWOup} - Q_{SH}$                                                     |
| $Q_{LHA}$     | Latent heat flux released by precipitation to atmosphere    | $\rho_0 L_V P$                                                                                    |
| $Q_{LHO}$     | Latent heat flux absorbed by evaporation from ocean/land    | $\rho_0 L_V E$                                                                                    |
| $Q_{LWAup}$   | Atmospheric longwave wave emission to space                 | $A + B T_A$                                                                                       |
| $Q_{LWAdown}$ | Atmospheric longwave wave emission to ground                | $\epsilon_{Ad} \sigma (273.15 + T_A)^4$                                                           |
| $Q_{LWOup}$   | Land/Ocean longwave wave emission to atmosphere             | $\epsilon \sigma (273.15 + T_{G/O})^4$                                                            |
| $Q_{SH}$      | Sensible heat flux exchange from ground/ocean to atmosphere | $\gamma (T_{G/O} - T_A)$                                                                          |
| $Q_{SSW}$     | Incoming solar short wave radiation TOA                     | $S_0 / 4 S(\vartheta)$                                                                            |
| $Q_{SSWA}$    | Atmospheric short wave absorption                           | $Q_{SSW} \frac{(1-\alpha)C_0(1+(1-C_0)\alpha_S)}{1-\alpha(1-C_0)^2\alpha_S}$                      |
| $Q_{SSWO}$    | Oceanic/land short wave absorption                          | $Q_{SSW} \frac{(1-\alpha)(1-C_0)(1-\alpha_S)}{1-\alpha(1-C_0)^2\alpha_S}$                         |
| $Q_{SSWR}$    | Short wave radiation reflected to space                     | $Q_{SSW} \left( \alpha + \frac{(1-\alpha)^2(1-C_0)^2\alpha_S}{1-\alpha(1-C_0)^2\alpha_S} \right)$ |
| $Q_{TA}$      | Atmospheric heat transport                                  | $C_A \nabla \cdot (K_A \nabla T_A)$                                                               |

Table 2: Summary of model parameterizations